

Research Statement

I am a macroeconomist with interests in labor markets and search. At the core of the study of the aggregate labor market is a desire to understand the causes of underutilization of resources in the aggregate economy, made especially tangible in the unemployment of workers. One of the biggest advances in our understanding of the labor market has come through the use of search theory, capturing the intuitive notion that the matching of workers and firms is costly in time and effort due to the different preferences, skills, and information possessed by workers and firms. My research focuses on the use of micro level data and computational modeling to study the interaction of search frictions and unemployment with various aspects of the macroeconomy. I address the fact that frictions in the labor market carry over into other dimensions of firms' and workers' decisions, such as a firm's incentive to utilize its current labor force, workers' participation in the labor market, and the decision to acquire or discharge debt.

Part-time Employment and Firm-level Labor Demand over the Business Cycle [Job Market Paper]

In addition to unemployment, workers may be underutilized through being employed part-time when they would desire full-time work. The population at work part-time for economic reasons (*PTE*) is countercyclical and fluctuates with the business cycle with similar volatility to the unemployment rate. The popular narrative of the involuntary part-time worker involves the recently unemployed who, for lack of better alternatives, has settled on part-time work. However, I show using data from the Current population survey and Survey of Income and Program Participants that the cyclical fluctuation in the population of part-time employed for economic reasons comes from the movement of workers between full-time employment and part-time rather than between part-time and unemployment. Not only is the fluctuation in *PTE* from within-employment, but it is largely due to within-job changes in hours.

Since fluctuations in *PTE* are driven by changes in the hours demanded of existing employees rather than changes in workers' job search or acceptance behavior, I model part-time work focusing on a firm's decision to hire, fire, or partially utilize its labor force in the face of firm-level and aggregate shocks. This requires a search model where firms are heterogeneous in size and productivity. The model produces part-time labor movements which are consistent with the evidence from worker flows in the CPS and match the relative volatility of unemployment and part-time for economic reasons over the business cycle. Part-time labor utilization by firms increases the volatility in vacancies and unemployment in the model relative to the model with only an extensive margin. Due to heterogeneity in firm size and productivity, the model has implications for the size and age distribution of firms and the growth rates of hours and employment.

The Role of Firm Size and Age on Employment and Hours Adjustments [Work in progress]

The role of firm characteristics such as size and age in the tradeoffs between hours and employment are highlighted in my model, but data on these adjustments at the firm-level is lacking. In ongoing work, I examine the cyclical dynamics of employment and hours across the distribution of firms. I utilize firm-level longitudinal micro data from the Longitudinal Business Database, Census of Manufactures, Annual Survey of Manufactures, and Quarterly Plant Capacity Utilization Survey at the U.S Census Bureau's Research Data Centers to measure the comovements and cyclical differentials in employment and hours adjustments by firm size, age, and productivity.

Matching Labor Flows in Search Models with Labor Force Participation [Working Paper]

Introducing a participation margin in a search model requires the use of large idiosyncratic shocks to workers' participation incentives to match the labor flows in the data. If we measure the participation transitions of workers outside of employment where search decisions are observable and apply this same transition process to employed workers, any search model will overstate the transition of workers out of employment to nonparticipation. Allowing the participation transition of workers to depend on their employment state fixes these flows, but this transition process is unobservable for employed workers. Taking advantage of the longer panel of the 1996 Survey of Income and Program Participants, I estimate the markov process for participation transitions of employed workers using their observed search behavior before and after an employment spell. The difference in the transition process measured for employed and nonemployed workers is consistent with an interpretation of attachment to the labor force. I build a directed search model with a labor force participation margin subject to employment-dependent shocks and show that it can match the labor market flows in US data.

Student Loans and Labor Market Outcomes [Working Paper] (with Chander S. Kochar)

The student loan market is the second largest source of household debt in the United States, with \$1.2 trillion in outstanding debt. Unlike other sources of unsecured credit, student loans cannot be discharged in bankruptcy. Using data on college graduates from the 1993/03 Baccalaureate and Beyond Longitudinal Study, we first identify that student loan debt has a significant negative effect on students' earnings after graduation. We show that the inability to discharge debt in bankruptcy is critical to produce this result within a simple search theoretic framework. We propose a richer model with student loan debt and a delinquency/default decision to study the effects of recent changes to student loan policies on the labor market and delinquency outcomes of college graduates.

Teaching Philosophy

My teaching experience at The University of Iowa has included working with graduate and undergraduate students as a teaching assistant, tutor, and lecturer. Each experience and teaching role has presented its own challenges and rewards, and provided me with a better understanding of the learning and teaching process at both the introductory and advanced level. I have high standards for both myself and my students, and this commitment to excellence is evidenced by my reception of the Economics Department's Outstanding Teaching Assistant Award.

Teaching is an integral part of the process of understanding any topic, and in my view is inexorably linked to research. In both cases, the end goal is similar: to persuade others of the merit of new ideas and provide the tools for their application. While in research, these ideas may be novel to the field and push the state of our collective understanding, in both cases the audience is previously uninitiated and must first be persuaded of the merit and importance of the material at hand. To me, this highlights a primary requirement of successful teaching, which is the need to engage the student and convince them of the usefulness of economics and the relevance of its methods in understanding their environment.

The value of motivation was made especially clear to me through lecturing in my own course in intermediate macroeconomics. My students were most engaged and motivated when they felt connected to topics through events and scenarios that directly affected and interested them. Showing students that they already face the tradeoffs in their models and placing them in the context of the decision maker helps them to realize the intuitiveness and usefulness of economic theories. The second aspect of motivation is to convince them of the significance of the material and how it applies to the world beyond their immediate scope and the topics in the classroom. Luckily, economics seeks the answer to some of the most important questions facing society and applies to almost every aspect of modern life. This has also been a benefit to my own research in helping me to focus on the bigger ideas and questions that my work seeks to address, and find ways to motivate their importance to a broader audience.

The challenge in teaching is often in knowing the right balance between focusing on providing the broad understanding and application of topics and imparting the technical proficiency and toolset to further analyze them. Understanding the needs and abilities of students is critical to know this balance. My experience tutoring individual students helped me to see the necessity as well as the challenges in addressing these two goals. The interaction of students with the teacher through assignments and in person is critical in knowing when one aspect is lacking, and I try to provide an accessible environment for questions and student input inside and outside of the classroom. Technology facilitates timely and frequent conversation with students through course discussions and forums, which I use to supplement lectures with novel topics or current events and to provide and receive feedback.

My experience working with other graduate students in the Analysis and Macro sequences gave me an appreciation for the role of teaching in facilitating my own learning and development. Guiding other students through the Ph.D. core was especially exciting, and through their discussions and

questions, helped me to arrive at a better understanding of both the tools of economics and its intuition. Through my experiences, I feel that I have improved both as a teacher and as an economist, and look forward to continuing to converse and engage with both my students and colleagues.

Table 1: **Teaching Evaluation Average Scores**

Course	Intermediate Macro	Grad Macro I	Grad Macro I	Grad Econ Analysis I	Grad Econ Analysis II	Principles of Micro
Term	Spring	Fall	Fall	Fall	Spring	Fall
Year	2015	2014	2013	2012	2012	2011
Instructor Evaluation Criteria	Mean of Responses: (1 =Strongly Disagree, 6 = Strongly Agree)					
Concepts are presented in a manner that helps me learn.	5.06	5.4	5.86	5.80	5.80	5.34
The instructor seems interested in teaching this course.	5.53	5.8	6.0	5.0	6.0	5.54
Help is available outside of class if I have questions.	5.71	6.0	6.0	5.4	6.0	5.72
This course is well planned and organized.	4.65	5.5	5.86	4.6	6.0	5.27
I would recommend a course taught by this instructor.	5.18	5.9	5.86	5.2	6.0	5.37
I learned a lot in this class.	4.82	5.3	5.86	5.4	5.8	5.12
Overall, this instructor is an effective teacher.	4.88	5.7	6.0	5.0	5.8	5.27
I am encouraged to apply new knowledge and skills.	4.88	5.6	6.0	5.6	6.0	5.15
Assignments contribute to my learning.	4.35	5.7	6.0	5.8	6.0	5.37
My critical thinking skills have improved because of this class.	4.65	5.3	5.86	5.4	6.0	4.88
The instructor communicates well.	5.18	6.0	6.0	6.0	6.0	5.50
The instructor understands my questions and comments.	5.44	5.9	6.0	6.0	6.0	5.59
Graded homework assignments were returned promptly.	5.69	5.9	6.0	6.0	6.0	5.66
I receive constructive criticism of written reports.	4.75	6.0	6.0	6.0	5.6	5.22
The instructor's oral comm. skills are adequate for this course.	5.44	6.0	6.0	6.0	6.0	5.59
Sample Size:	$N = 17$	$N = 10$	$N = 7$	$N = 5$	$N = 5$	$N = 41$